



Antimicrobial resistance in Indian isolates of non typhoidal *Salmonella* of livestock, poultry and environmental origin from 1990 to 2017

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ABSTRACT

A retrospective antimicrobial resistance study of nontyphoidal *Salmonella enterica* isolates from India during 1990–2017 was conducted to study the microbial susceptibility to antibiotics. A total of 271 *Salmonella enterica* isolates from poultry (n = 146), farm animals (n = 55) and environmental sources (n = 70) were tested for susceptibility using 15 antimicrobial drugs. The drug classes include aminoglycosides, phenicols, cephalosporins, penicillins, carbapenems, fluoroquinolones, and sulphonamide-trimethoprim. Study revealed that overall, 133 (49.08%) of 271 isolates were resistant to ≥ 1 antimicrobial drugs and 81 (29.89%) out of 271 isolates were multidrug resistant (resistance to ≥ 3 drugs). Majority (68.96%) of Typhimurium serovars (n = 87) were susceptible to all antibiotics tested, whereas only 5% Kentucky serovars (n = 40) were pan susceptible. All the drugs revealed decreasing trend of susceptibility from 1990 towards 2017 except cephalosporins and carbapenems. Statistical analysis of association between time period and antimicrobial resistance revealed a significance of < 0.05 . Molecular detection of genetic determinants associated with antimicrobial resistance revealed the presence of genes like class I integrons, *sul1*, *sul2*, *catIII*, *cmlA*, *dfrA*, *bla_{TEM}*, *bla_{AmpC}* in the resistant isolates. Furthermore, plasmid mediated quinolone resistant determinants like *qnrD* and *qnrS* were also reported in the current study.

1. Introduction

Non typhoidal *Salmonella* is one of the principal causes of food poisoning worldwide with an estimated annual incidence of 1.3 billion cases and 3 million deaths each year [1]. Non typhoidal fever, which is caused mainly by *Salmonella* Typhimurium and *Salmonella* Enteritidis, continues to be a major problem in developing countries. Food animals act as major reservoirs of non typhoidal *Salmonella* and it is one of the important food-borne zoonotic pathogens. In animals, non typhoidal *Salmonella* cause abortions (rare), diarrhoea, reduced feed conversion in cattle, gastro-enteritis, septicaemia, abortion in small ruminants, fever, edema, diarrhea, hypoxia in pigs. In poultry, they are associated with Pullorum disease, fowl typhoid, paratyphoid [2]. For many years, chloramphenicol, ampicillin, and trimethoprim-sulphamethoxazole (cotrimoxazole) were the drugs of choice for treatment of non-typhoidal salmonellosis. In recent years, the emergence of multi-drug resistant (MDR) isolates has become a matter of concern. A typical

example is the *S. Typhimurium* definitive type (DT) 104 which imparts resistance to ≥ 5 antibiotics namely ampicillin, chloramphenicol, florfenicol, streptomycin, sulfonamides, tetracyclines and trimethoprim from animal sources [3]. Although fluoroquinolones, such as ciprofloxacin and ofloxacin, and extended spectrum cephalosporins, such as ceftriaxone and cefotaxime, have proved to be effective alternatives, resistance to these agents has also emerged [4]. In the 2010 s, emergence of *S. Kentucky* ST-198-X1 strain resistant to amoxicillin, gentamicin, sulphonamides and ciprofloxacin were noticed in south-east Asian countries followed by its spread worldwide [5]. However, there was lack of knowledge about the level of antimicrobial resistance in Indian NTS isolates. Therefore, the current study was conducted to study the antimicrobial resistance in Indian isolates of NTS from poultry, farm animals and environment over a period of 27 years.

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2. Materials and methods

2.1. Bacterial strains

A total of 271 cultures of *Salmonella enterica* isolated from different parts of India from time period ranging from 1990 to 2017 and maintained in nutrient slants in the culture repository of National Salmonella Centre, Division of Bacteriology and Mycology, Indian Veterinary Research Institute, Izatnagar were used for the study. The cultures in nutrient agar were revived by enrichment in brain heart infusion (BHI) broth followed by incubation at 37° C. The enriched cultures were streaked on Hektoen enteric (HE) agar and the plates were incubated at 37°C for 18–24 h. The biochemical characterization was done using Triple Sugar Iron (TSI) agar and urease broth. The molecular characterization of *Salmonella enterica* isolates were done by PCR targeting the *invA* gene. The details of the primers were given in the [supplementary table](#).

2.2. Antibiotic sensitivity test

The isolates (n = 271) were tested for their susceptibility pattern using a panel of 15 different antibiotics (BD BBL Sensi-disc, Becton Dickinson and Company, New Jersey, USA) namely ampicillin 10 µg, tetracycline 30 µg, streptomycin 10 µg, chloramphenicol 30 µg, colistin 10 µg, amoxicillin/clavulanic acid 20/10 µg, imipenem 10 µg, meropenem 10 µg, colistin 10 µg, amoxicillin/clavulanic acid 20/10 µg, imipenem 10 µg, meropenem 10 µg, ceftazidime 30 µg, sulphamethoxazole/trimethoprim 23.75/1.25 µg, ceftazidime/clavulanic acid 30/10 µg, cefotaxime 30 µg, cefotaxime /clavulanic acid 30/10 µg, ciprofloxacin 5 µg, levofloxacin 5 µg and nalidixic acid 30 µg. Antibiotic susceptibility testing was performed following the Kirby-Bauer disc diffusion method [6] as per the Clinical and Laboratory Standards Institute guidelines [7]. Mueller Hinton Agar (Hi-media laboratories, India) plates were used and incubated aerobically at 37° C for 24 h. *Escherichia coli* ATCC 25922 was used as a quality control strain throughout the test procedure. The minimum inhibitory concentration (MIC) values for ampicillin (0.016–256 µg), tetracycline (0.016–256 µg), streptomycin (0.016–256 µg) and co-trimoxazole (0.002–32 µg) were determined using E-test strips (Hi-media laboratories, India) in representative isolates. Thirty representative poultry isolates which were resistant to fluoroquinolones were tested for MIC determination using ciprofloxacin (0.016–256 µg) and levofloxacin (0.002–32 µg) E-test strips (Hi-media laboratories, India).

2.3. Molecular detection of antimicrobial resistance

The previously extracted snap chilled DNA from selected isolates (n = 30) was further examined for the presence of various antimicrobial resistance genes. The list of primers for detection of antimicrobial resistant genes was given in [supplementary table](#).

2.4. Statistical analysis

Pearson's chi-square (χ^2) test was used to determine the association of various factors such as serovar, sources of isolates and year with antimicrobial resistance. significant differences ($P < 0.05$) in frequency of antimicrobial resistance profiles among NTS isolates obtained from different sources over a time period.

3. Results

3.1. Revival and molecular detection of NTS isolates

The revived *Salmonella enterica* isolates in HE Agar produced characteristic greenish or bluish green colonies with black precipitate at the centre. The TSI test revealed red slant, yellow butt with black

precipitation indicative of *Salmonella enterica* isolates. The urease broth remained colourless indicating absence of urease production. The PCR reaction targeting *invA* gene produced specific amplification at 284 bp on running in agarose gel electrophoresis in 271 NTS isolates.

3.2. Antibiotic sensitivity test

3.2.1. Antimicrobial resistance in NTS isolated from various sources

Overall, 133 (49.08%) of 271 isolates were resistant to ≥ 1 antimicrobial drugs. The most common resistance phenotype were to older drugs like nalidixic acid (40.96%), followed by tetracycline (29.89%), ampicillin (26.2%), streptomycin (22.14%), sulphadiazine-trimethoprim (17.71%), ciprofloxacin (16.97%), levofloxacin (15.13%), amoxicillin-clavulanic acid (9.96%) and chloramphenicol (4.06%). Overall, 81 (29.89%) out of 271 isolates were multidrug resistant (resistance to ≥ 3 drugs). When analyzed by source, 63.64% of farm animal isolates were pan susceptible, followed by 58.57% environmental isolates and 41.78% of poultry isolates. Multidrug resistance was observed in 39.73% poultry isolates, 21.81% farm animal isolates and 15.71% environmental isolates. Majority of the poultry isolates were resistant to older drugs such as nalidixic acid (50.68%), followed by tetracycline (36.3%), ampicillin (30.82%) and streptomycin (29.45%). About 32% of farm animal isolates were resistant to nalidixic acid followed by ampicillin and sulphadiazine (25.45%). Antimicrobial drug resistance was less noticed in the environmental samples with 27.14% resistance to nalidixic acid followed by streptomycin (18.57%) and sulphadiazine-trimethoprim (17.14%). Statistical analysis of relationship between sources of the isolates versus antimicrobial resistance revealed a significance of < 0.005 which is depicted in [Table 1](#).

3.2.2. Serovar-wise antimicrobial resistance in NTS isolates

Typhimurium (n = 87) and Kentucky (n = 40) were the major serovars present in the study and characteristic susceptibility profiles appeared. Therefore, they both were considered as separate serovars whereas all others (Virchow-13; Enteritidis-12; Paratyphi B-8; *Salmonella* rough-17; *Salmonella* untypeable-94) were categorized into other serovars (n = 144). Majority (68.96%) of Typhimurium serovars were susceptible to all antibiotics tested, followed by other serovars (47.91%) whereas only 5% Kentucky serovars were pan susceptible. Resistance to 1–2 antimicrobial drugs were observed in 25.69% of other serovars, 18.39% of Typhimurium and 2.5% Kentucky isolates. Multidrug resistance was much higher in Kentucky isolates (92.5%), 26.39% in other serovars and 12.64% in Typhimurium isolates. When Typhimurium resistance to individual drugs was observed, nalidixic acid was comparatively resistant (21.84%) followed by streptomycin (13.79%) whereas the serovars were highly sensitive to chloramphenicol (100% sensitive). In Kentucky isolates, 90% resistance was observed for nalidixic acid, followed by tetracycline (87.5%), ampicillin (77.5%), ciprofloxacin (75%) and levofloxacin (75%). Nalidixic acid resistance was found to be common (34.72%) in other serovars followed by tetracycline (25.69%), sulphanamide-trimethoprim (21.53%), ampicillin (20.83%),

Table 1

Resistance profile of *Salmonella enterica* isolates from different sources.

S. No	Antibiotic	Poultry (n = 146)	Farm animals (n = 55)	Environment (n = 70)	df	Chi-Square
1	Overall susceptible	61	35	41	8	38.454***
2	Resistant to 1–2 antibiotics	27	8	18		
3	Resistant to ≥ 3 antibiotics	58	12	11		

*** $P < 0.001$

amoxicillin-clavulinate (4.16%) and levofloxacin (4.16%). Statistical analysis of the relationship between serovars and antimicrobial resistance frequency revealed a significance of < 0.001 which is outlined in Table 2.

3.2.3. Decade wise AMR in NTS isolates

Over the period from 1990 to 2000, antimicrobial drug susceptibility of ampicillin was 96.88% which decreased to 80.81% in 2001–2010 and to 64.28% in 2011–2017 isolates. The antimicrobial susceptibility pattern of chloramphenicol was 100%, 98.99% and 95.71% in 1990–2000, 2001–2010 and 2011–2017 isolates, respectively. However, in case of old drugs like streptomycin, the susceptibility has reduced from 93.75% in 1990–2000–84.85% in 2001–2010 and further to 70% in isolates from 2011–2017. However, the susceptibility of sulphadiazine-trimethoprim was 100% in 1990–2000 whereas 80% in cultures isolated from 2001–2010 and 2011–2017. The susceptibility of tetracycline was 96.88% in 1990–2000 which further reduced to 82.83% in 2001–2010 and 57.14% during the years 2011–2017. There were 90.63% nalidixic acid susceptible strains from 1990–2000, which decreased to 71.71% in 2001–2010 and further reduction to 46.43% in 2011–2017. The isolates were completely susceptible to ciprofloxacin in 1990–2000, whereas the susceptibility pattern reduced to 94.95% in 2001–2010 and 71.43% in 2011–2017. The susceptibility pattern of levofloxacin to NTS isolates were 100%, 97.98% and 73.57% during the period 1990–2000, 2001–2010 and 2011–2017, respectively. The susceptibility pattern of amoxicillin-clavulanic acid was 100%, 96.97% and 82.14% respectively during the period 1990–2000, 2001–2010 and 2011–2017. Statistical analysis of association between time period and antimicrobial resistance revealed a significance of < 0.05 which was tabulated in Table 3.

3.3. Minimum inhibitory concentration of selected NTS isolates

A few representative isolates were selected for determination of minimum inhibitory concentration of antimicrobial drug for the resistant isolates. For the drug ampicillin, 26 isolates (86.67%) were resistant by (Antibiotic Sensitivity Test) ABST whereas 22 isolates (73.33%) were resistant by E-test. Majority of the isolates showed $> 256 \mu\text{g/L}$ MIC for ampicillin. Out of the 30 isolates, 18 (60%) were resistant by ABST and 17 isolates (56.67%) were resistant by E-strip test to streptomycin. When the thirty isolates were tested for tetracycline resistance, 26 (86.67%) were observed as resistant by ABST and 23 (76.67%) were resistant by E-strip test. The MIC of the resistant isolates was observed to be $> 256 \mu\text{g/L}$ to tetracycline and the susceptible isolates were showing MIC of $\leq 1.5 \mu\text{g/L}$. When tested for co-trimoxazole resistance, 22 (73.33%) out of 30 isolates were positive by ABST and 15 (50%) out of 30 were positive by E-strip test. The MIC range of the resistant isolates was $\geq 32 \mu\text{g/mL}$ and the susceptible isolates were from 0.12 to 0.012 $\mu\text{g/mL}$.

For the drug ciprofloxacin, 25 isolates (83.33%) were resistant by ABST and 29 isolates (96.67%) were observed as resistant by E-strip test. The susceptible isolate has a MIC of 1 $\mu\text{g/mL}$ whereas the MIC of resistant isolates was in the range of 16–192 $\mu\text{g/mL}$. When tested for levofloxacin resistance, 26 isolates (86.67%) were observed as resistant by ABST and 25 isolates (83.37%) isolates were resistant by E-strip test. An MIC range of 0.064 – 1.5 $\mu\text{g/mL}$ was noticed in susceptible isolates and $\geq 32 \mu\text{g/mL}$ was noticed for resistant isolates. The tabular

representation was given in Table 4.

3.4. Molecular detection of antimicrobial resistance

Out of the 30 isolates, *tetA*, *tetB*, *tetG* genes was present in 17, 5 and 0 isolates, respectively. Molecular detection of chloramphenicol resistance revealed the presence of *cmlA* gene in 5 isolates, *catA3* gene was positive in 5 isolates whereas none of the isolates possessed *catII* gene. It is of particular importance, that molecular determinants of chloramphenicol resistance were detected in the isolates, phenotypically, very few isolates exhibited chloramphenicol resistance (4.05%). The gene, *sulI* was present in 11 isolates whereas 3 isolates were positive by *sul2* gene and three isolates were positive for *dfrA* gene encoding trimethoprim resistance. The beta lactamase gene, *bla_{TEM}* was present in 9 isolates whereas none of the isolates were positive for *pse* gene. Among the 30 resistant isolates, 16 were positive for Class I integron genes whereas none of them were positive for class II integrons. Sanger di-deoxy sequencing of representative class I integrons revealed the presence of antimicrobial resistance genes such as *tetA*, *aad* and *dfrA5*. The *blaAmpC* (AmpC beta lactamase) gene was amplified in 2 isolates whereas the genes for ESBL and carbapenem resistance like *blaCTXM*, *blaIMP*, *blaVIM*, *blaOXA*, *blaNDM*, *blaKPC*, *blaSHV* were absent. Multiplex PCR of fluoroquinolone resistant isolates revealed four isolates were positive for *qnrD* gene and two isolates were positive for *qnrS* gene. The *strA* (aminoglycoside 3' adenylyl transferase) gene, erythromycin resistant gene, *erm* and colistin resistant genes namely *mcr1* and *mcr2* were absent in the screened isolates. The nucleotide sequences were allotted accession numbers viz. MT521879-MT521883, MT555122-MT555123, MT370467-MT370471. Table 4 and 5 discussed the phenotypic, MIC values and molecular determinants found in selected isolates.

4. Discussion

In the present study, revival, molecular characterization and antimicrobial resistance of NTS from livestock, poultry and environmental origin from 1990 to 2017 were done. The gene, *invA* was used for molecular characterization of NTS isolates. As it is present in only *Salmonella*, it can be called as the characteristic molecular marker to identify the genus and helps in differentiating from other members of Enterobacteriaceae family [8]. Our study revealed resistance to older antibiotic classes such as quinolones, tetracyclines, aminoglycosides, sulphonamides and penicillins whereas emerging resistance to fluoroquinolone antimicrobials were noticed. The increased level of resistance to traditional antimicrobials viz. nalidixic acid, sulphonamides, streptomycin, tetracycline, ampicillin is evident in various serovars of NTS isolated worldwide [9,10]. It was reported that ASSuT (ampicillin, streptomycin, sulfonamides, and tetracycline) and ACSSuT patterns were noticed majority of *S. Typhimurium* isolates (43%) [11] whereas in the current study, ASSuTNx and ASTNxCipLvx patterns were observed. However, antimicrobial susceptibility patterns in NTS vary with the prevalence of a serovar, its distribution, antibiotic use in human, veterinary medicine and animal husbandry. The pattern also revealed that fluoroquinolone resistance was increasing trend in Indian NTS isolates than third generation cephalosporins and carbapenem resistance. This may be due to increased prevalence of Kentucky serovars after 2010. A similar observation has been observed during surveillance by European Union

Table 2
Resistance profile of different *Salmonella enterica* serovars.

S.No	Antibiotic	Typhimurium (n = 87)	Kentucky (n = 40)	Others (n = 144)	df	Chi-Square
1	Overall susceptible	60	2	69	8	120.364***
2	Resistant to 1–2 antibiotics	16	1	37		
3	Resistant to ≥ 3 antibiotics	11	37	30		

(Others include Virchow-13; Enteritidis-12; Paratyphi B-8; *Salmonella* rough-17; *Salmonella* untypeable-94)

*** $P < 0.001$

Table 3Resistance profile of *Salmonella enterica* isolates from different decades.

S.No	Antibiotic	1990–2000 (n = 30)	2001–2010 (n = 100)	2011–2017 (n = 141)	df	Chi-Square
1	Overall susceptible	23	60	54	4	29.431***
2	Resistant to 1–2 antibiotics	5	23	25		
3	Resistant to ≥ 3 antibiotics	2	17	62		

*** P < 0.001

Table 4Antibiotic resistant pattern, Minimum inhibitory concentration (MIC) and genetic determinants of *Salmonella enterica*.

S. No	Serovar	Year	Source	Antibiotic resistant pattern	Antibiotic resistant determinants	AMP	STR	TE	COT	C
1	Salmonella untypeable	2004	Poultry	AMP, COT, TE, NX	TetA, sul1, cmlA, blaTEM, class I integrons	> 256	4	> 256	> 32	8
2	Virchow	2017	Poultry	AMP, COT, TE, NX, CIP, LVX	TetB, catIII, class I integrons	1	12	> 256	> 32	64
3	Kentucky	2017	Poultry	AMP, AMC, TE, STR, NX, CIP, LVX	TetA, sul1, blaTEM, class I integrons	> 256	48	> 256	0.19	4
4	Kentucky	2017	Poultry	AMP, AMC, TE, STR, NX, CIP, LVX	Sul1, class I integrons	> 256	64	> 256	0.19	6
5	Kentucky	2017	Poultry	AMP, C, COT, TE, STR, NX, CIP, LVX	TetB, catIII, qnrD	12	12	> 256	0.125	24
6	Salmonella Rough	2002	Goat	AMP, COT, TE, NX	TetB, sul1, cmlA, catIII, class I integron	1.5	24	1.5	0.047	4
7	Salmonella rough	2017	Pig	AMP, STR, TET, SUL, NX	TetA, class I integron	> 256	24	> 256	> 32	32
8	Enteritidis	2014	Human	NX, LVX	Class I integron	1	4	3	0.064	3
9	Typhimurium	2015	poultry	COT, TE, STR, NX	TetA, cmlA	0.75	32	0.33	0.047	4
10	Salmonella untypeable	2005	Poultry	AMP, C, STR, NX	Class I integrons	1.5	16	1.5	0.023	4
11	Virchow	2017	Pig	AMP, C, TE, COT, NX, CIP	Sul1, dfrA, blaTEM, class I integrons	> 256	8	> 256	> 32	32
12	Virchow	2017	Pig	AMP, C, COT, STR, TE, NX	TetB, sul1, dfrA, catIII	> 16	16	> 256	0.125	32
13	Kentucky	2017	Poultry	AMP, COT, TE, STR, NX, CIP, LVX	tetA	> 256	8	> 256	0.064	3
14	Kentucky	2017	Poultry	COT, TE, STR, NX, CIP, LVX	TetB, sul1, cmlA	1	24	> 256	0.047	24
15	Kentucky	2017	Poultry	AMP, AMC, TE, STR, NX, CIP, LVX	TetA, Class I integrons	> 256	12	> 256	0.047	3
16	Rough salmonella	2017	Poultry	AMP, C, COT, TE, NX	TetA, sul1, dfrA, blaTEM, class I integrons	> 256	12	> 256	> 32	32
17	Enteritidis	2013	Poultry	AMP, AMC, C, STR, TE, NX	TetA, sul1, blaTEM, class I integron	> 256	48	> 256	0.125	12
18	Salmonella untypeable	2010	Environment	AMP, AMC, COT, STR, TE, NX, CIP, LVX	TetA, sul1, sul2	> 256	24	> 256	> 32	4
19	Typhimurium	2010	Poultry	AMP, COT, STR	Sul2	4	16	> 256	0.25	32
20	Salmonella untypeable	2005	Poultry	NX	–	1	24	1.5	0.012	4
21	Salmonella untypeable	2002	Goat	AMP, COT, TE, NX	TetA, sul1, cmlA, Class I integron	> 256	4	> 256	> 32	4
22	Typhimurium	2017	Poultry	NX	TetA, sul2, blaTEM, blaAmpC, qnrS	> 256	48	> 256	> 32	4
23	Salmonella untypeable	2002	Calf	AMP, COT, TE, NX	TetA, sul1, class I integrons	> 256	6	> 256	> 32	6
24	Enteritidis	2013	Poultry	AMC, C, TE, STR, NX	TetA, sul1, sul2, blaTEM, blaAmpC	> 256	32	> 256	> 32	4
25	Paratyphi B	2001	Dog	AMP, COT, TE	TetA, cmlA, class I integrons	> 256	3	> 256	> 32	4
26	Kentucky	2017	Poultry	AMP, AMC, COT, TE, STR, NX, CIP, LVX	TetA	> 256	24	> 256	> 32	64
27	Virchow	2004	Environment	AMP, COT, TE, STR, NX	TetA, sul1, cmlA, blaTEM, class I integron	> 256	12	1.5	> 32	6
28	Salmonella rough	2015	Environment	AMP, AMC, TE, STR, NX, CIP, LVX	TetA, sul1, class I integron	> 256	32	> 256	0.094	3
29	Salmonella untypeable	2002	Pig	AMP, COT, TE, NX	TetA, sul1, class I integrons	> 256	2	> 256	> 32	4

AMP- Ampicillin, AMC-Amoxicillin-clavulanic acid, COT-Co-trimoxazole, TE-Tetracycline, NX-Nalidixic acid, CIP-Ciprofloxacin, LVX-Levofloxacin, STR-Streptomycin

whereas NTS isolates from the United States were susceptible to fluoroquinolones but resistant to third generation cephalosporins [9]. Chloramphenicol resistance was least observed in the study due to less usage of the drug owing to its toxicity. Therefore, other classes of drugs, mainly fluoroquinolones and cephalosporins came into use. In our study, resistance to fluoroquinolones such as ciprofloxacin and levofloxacin were observed in isolates isolated after 2010. This is of particular concern, as fluoroquinolones are the drug of choice during invasive NTS infections [12].

The isolates were completely susceptible to imipenem, meropenem, ceftazidime, cefotaxime and their clavulanate counterparts. In contrast, 5% resistance to ceftriaxone in human NTS isolates from south India [13] and 41.43% NTS isolates from chicken meat shops across north India were reported to be co-resistant to ciprofloxacin and cefotaxime [14]. Our reports were consistent with previous studies conducted in other countries which reported that poultry isolates were more resistant than farm animal isolates [15]. Various authors have reported the emergence of *S. Kentucky* in south-east Asian countries after 2010 s and

Table 5Antibiotic resistant pattern, Minimum inhibitory concentration (MIC) and genetic determinants of *Salmonella enterica* in poultry against quinolones.

S.no	Source	Year	Antibiotic resistance pattern			MIC		qnrD	qnrS
			NX	CIP	LVX	CIP	LVX		
1	Poultry	2017	R	R	R	16	> 32	–	–
2	Poultry	2017	R	R	R	16	0.032	–	–
3	Poultry	2017	R	R	R	16	> 32	–	–
4	Poultry	2017	R	S	R	16	> 32	–	–
5	Poultry	2017	R	R	R	24	> 32	–	–
6	Poultry	2017	R	R	R	16	> 32	–	–
7	Poultry	2017	R	R	R	30	1.5	+	–
8	Poultry	2017				24	> 32	–	–
9	Poultry	2017	R	R	S	192	> 32	–	–
10	Poultry	2017	R	R	R	16	24	–	–
11	Poultry	2017	R	R	R	16	24	–	–
12	Poultry	2017	R	R	R	24	24	+	–
13	Poultry	2017	R	S	S		1.5	–	+
14	Poultry	2017	R	S	R	24	> 32	–	–
15	Poultry	2017	S	S	R	1	0.75	–	–
16	Poultry	2017	R	S	R	16	> 32	–	–
17	Poultry	2017	R	R	S	24	24	–	–
18	Poultry	2017	R	R	R	30	0.064	–	–
19	Poultry	2017	R	R	R	128	> 32	–	–
20	Poultry	2017	R	R	R	24	> 32	–	–
21	Poultry	2017	R	R	R	16	> 32	–	–
22	Poultry	2017	R	R	R	32	> 32	–	–
23	Poultry	2017	R	R	R	16	> 32	–	–
24	Poultry	2017	R	R	R	16	> 32	–	–
25	Poultry	2017	R	R	R	48	> 32	+	–
26	Poultry	2017	R	R	R	48	> 32	–	–
27	Poultry	2017	R	R	R	32	> 32	+	–
28	Poultry	2017	R	R	R	32	> 32	–	–

their resistance to fluoroquinolones and third generation cephalosporins [14,16].

Most of these antibiotic resistance genes are not organism specific and acquired through horizontal gene transmission. Therefore, they may get lost in the absence of antibiotic selection pressure or withdrawal of the particular antibiotic from use in animal husbandry. However, the menace may still persist, if the genes get incorporated into *Salmonella* genomic islands (SGI, SGII) such as integrons. Class 1 integrons are associated with a variety of resistance gene cassettes such as *aadA* resistance determinants, encoding streptomycin-spectinomycin resistance. In addition, genes for trimethoprim and beta lactam resistance determinants are also detected frequently [17–19]. Sanger dideoxy sequencing of class I integrons revealed resistant determinants like dihydrofolate reductase (*dhfrA*) and aminoglycoside-N-acetyl transferase (*aad*) which impart resistance against drugs like trimethoprim and aminoglycosides, respectively.

Plasmid mediated quinolone resistance (PMQR) is an emerging challenge in transmission of antibiotic resistance to human food chain. In the current study, PMQR determinants such as *qnrS* and *qnrD* were observed. An international survey conducted in 13 European countries revealed *qnrS* in 10% and *qnrD* gene has been identified in 1.81% *Salmonella* isolates in human and animal isolates from different European countries [20]. To our knowledge, *qnrD* genes were reported for the first time in Indian NTS isolates.

5. Conclusion

The current study confirmed resistance to older drugs like nalidixic acid, tetracycline, ampicillin and streptomycin by conventional and molecular methods. Although, resistance to critically important antibiotics like cephalosporins, carbapenems and colistin were not reported, the emergence of fluoroquinolone resistance possesses a threat to poultry, farm animal production and public health. Antibiotics are generally not recommended for control of salmonellosis in positive herds or flocks. In spite, in the recent decades, there was decrease in antimicrobial drug susceptibility to fluoroquinolones and third

generation cephalosporins in South Asian countries. Therefore, the current study will help to understand the antimicrobial drug susceptibility of non typhoidal *Salmonella* isolated from various sources such as poultry, farm animals and environment over a period of time across the Indian subcontinent. The resistance genes found in the current study could help in epidemiological tracing of genes between bacterial species, prediction of future outbreaks in different regions and to formulate treatment strategies.

CRedit authorship contribution statement

Sophia Inbaraj and Ravi Kant Agrawal: designed the study, collected data, conducted the experimental work, analysed data and wrote the manuscript. **Prasad Thomas and Agarwal, R.K:** Design of the project, and writing and editing the manuscript. **Chandra Mohan and Med Ram Verma:** Performed data analysis, Edition. **Pallab Chaudhuri:** Edited the final manuscript. All authors read and approved the final manuscript.

Conflict of interest

The authors declare that they have no conflict of interest.

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Appendix A. Supporting information

associated with this article can be found in the online version at [doi:10.1016/j.cimid.2021.101719](https://doi.org/10.1016/j.cimid.2021.101719).

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